520.681: Perception for Embodied AI
Homework 1 - Coding
Due: Feb 18, 2026, 11:59pm ET

Instructions

- This is an individual homework and the coding part is worth 100 points
- You must submit your solutions on Gradescope.
- Please complete this homework with Python 3. To run this homework successfully, you will need the following Python packages:
  - numpy
  - matplotlib
  - opencv-python
- Start early! Please post your questions on Canvas or come to office hours.

Submission

- You will submit all python files you completed as .py files to Gradescope

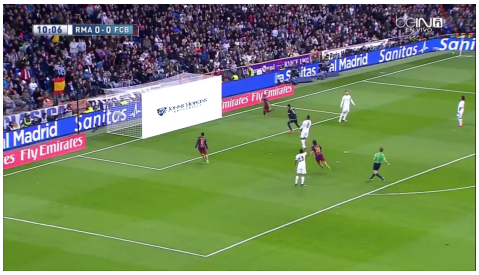
1 AR on the Pitch (100 pts)

1.1 Introduction

In this programming assignment, we will use homographies to project an image onto a scene. To demonstrate this, we will project our logo onto the goal during a football match. We have provided images from a video sequence of a football match, the corners of the goal in each image, and an image of the Johns Hopkins logo. For each frame, you will compute the homography between the Hopkins logo and the goal, and then warp

points on the Hopkins logo to the goal points (vertical plane) to generate a projection of the logo onto the video frame (e.g. Figure 1).

(a) Original image from video



(b) Image with projected Hopkins logo

Figure 1: Example of logo projection

1.2 Technical Details

The Python script **project_logo.py** will be the main script to run this assignment. In this script, we provide you the following:

- A sequence of images onto which you will project a logo image.
- The corners in each video frame that the logo should project onto.
- The logo image itself.

Your goal will be to complete the functions **est_homography** and **warp_pts** in their respective .py files. **est_homography** estimates the homography that maps the video image points onto the logo points (i.e. $\mathbf{x}_{logo} \sim \mathbf{x}_{video}$), and **warp_pts** then warps the sample points according to this homography, returning the warped positions of the points (that is, the corresponding points in the logo image). We then use these correspondences to copy the points from the logo into the video image. Once you finish these two functions, change your current directory to the directory of "Part_1", and run "python project_logo.py". It will use the two functions you completed to project logo, visualize them in a video, and save sampled results.

1.3 Homography Estimation

To project one image patch onto another, we need, for each point inside the goal in the video frame, to find the corresponding point from the logo image to copy over. In other words, we need to calculate the homography between the two image patches. This homography is a 3x3 matrix that satisfies the following:

$$\mathbf{x}_{logo} \sim H\mathbf{x}_{video} \tag{1}$$

Or, equivalently:

$$\lambda \mathbf{x}_{logo} = H \mathbf{x}_{video} \tag{2}$$

Where $\mathbf{x}_{logo}$ and $\mathbf{x}_{video}$ are homogeneous image coordinates from each patch and λ is some scaling constant. To calculate the homography needed for this projection, we provide, for each image, the corners of the patches that we would like you to warp between in each image. You must calculate the homography using the provided corner points and the technique covered in the lectures and Appendix A. You can then warp each image point using H to find its corresponding point in the logo (note that the homography equation is estimated up to a scalar, so you will need to divide $H\mathbf{x}_{image}$ by the third term, which is λ), and then return the set of corresponding points as a matrix.

1.4 Inverse Warping

You may be wondering why we are calculating the projection from the **video frames** to the **logo image**, when we want to project the logo image onto the video frames. We do this because, if we compute the inverse homography, and project all the logo points into the video frame, we will most likely have the case where multiple logo points project to one video frame pixel (due to rounding of the pixels), while other pixels may have no logo points at all. This results in "holes" in the video frame where no logo points are mapped. To avoid this, we calculate the projection from video frame points to logo points to guarantee that every video frame gets a point from the logo.

We can then replace every point in the video frame ($\mathbf{x}_{video}$) with the corresponding point in the logo ($\mathbf{x}_{logo}$) using the correspondences ($\mathbf{x}_{image}, \mathbf{x}_{logo}$). This is already done for you in **inverse_warping**, and you should not need to change it.

1.5 Files to complete and submit

1. **est_homography.py** (50 pts)
   This function is responsible for computing the homography given correspondence

2. **warp_pts.py** (50 pts)
   This function is responsible for computing the warped points given correspondence and a set of sample points

1.6 Visualizing Results

To play the projected images as a video, run the **project_logo.py** script. First open a terminal window and change the current directory to the directory of "Part_1". Then run command:

python project_logo.py

You can also use an IDE to run the script. If you would like to play with your own data, you can edit **project_logo** and generate your own video with a set of points.

1.7 Bonus Question (20 pts)

Record a short video (**approximately 5 seconds**) of a building on campus and select a rectangular planar region that you can track across time, such as a window or door. While recording, make sure the chosen rectangular patch moves **at least half of the image width** from the beginning to the end of the clip. Next, downsample the video uniformly to obtain **exactly 30 frames** in total. For each of these 30 frames, locate the same physical rectangular plane and manually click its four corner points; you may use AI tools to help you build an interface for efficient annotation, but the corner locations must be **provided by manual clicks**. Using the resulting corner correspondences, compute the per-frame warp (for example, via a homography), generate the warped images, and then reassemble the warped image sequence into a video using any tool of your choice, such as OpenCV and FFmpeg. Finally, ensure that the submitted video file is **under 10 MB** and follow these instructions closely, since **no bonus points will be given otherwise**.

2 Appendix A: Calculating Homographies

As we learned in the lectures, a homography H maps a set of points $\mathbf{x} = \begin{pmatrix} x & y & 1 \end{pmatrix}^T$ to another set of points $\mathbf{x}' = \begin{pmatrix} x' & y' & 1 \end{pmatrix}$ up to a scalar:

$$\mathbf{x}' \sim H\mathbf{x} \tag{3}$$

$$\lambda \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{4}$$

$$\lambda x' = h_{11}x + h_{12}y + h_{13} \tag{5}$$

$$\lambda y' = h_{21}x + h_{22}y + h_{23} \tag{6}$$

$$\lambda = h_{31}x + h_{32}y + h_{33} \tag{7}$$

In order to recover x' and y', we can divide equations (5) and (6) by (7):

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}} \tag{8}$$

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}} \tag{9}$$

Rearranging the terms above, we can get a set of equations that is linear in the terms of H:

$$-h_{11}x - h_{12}y - h_{13} + h_{31}xx' + h_{32}yx' + h_{33}x' = 0 \tag{10}$$
$$-h_{21}x - h_{22}y - h_{23} + h_{31}xy' + h_{32}yy' + h_{33}y' = 0 \tag{11}$$

Finally, we can write the above as a matrix equation:

$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} h = 0 \tag{12}$$

Where:

$$a_x = \begin{pmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \end{pmatrix} \tag{13}$$
$$a_y = \begin{pmatrix} 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \end{pmatrix} \tag{14}$$
$$h = \begin{pmatrix} h_{11} & h_{12} & h_{13} & h_{21} & h_{22} & h_{23} & h_{31} & h_{32} & h_{33} \end{pmatrix}^T \tag{15}$$

Our matrix H has 8 degrees of freedom, and so, as each point gives 2 sets of equations, we will need 4 points to solve for h uniquely. So, given four points (such as the corners provided for this assignment), we can generate vectors a_x and a_y for each, and concatenate them together:

$$A = \begin{pmatrix} a_{x,1} \\ a_{y_1} \\ \vdots \\ a_{x,n} \\ a_{y,n} \end{pmatrix} \tag{16}$$

Our problem is now:

$$Ah = 0 \tag{17}$$

As A is a 8x9 matrix, there is a unique null space. Normally, we can use MATLAB's **null** function, however, due to noise in our measurements, there may not be an h such that Ah is exactly 0. Instead, we have, for some small $\vec{\epsilon}$:

$$Ah = \vec{\epsilon} \tag{18}$$

To resolve this issue, we can find the vector h that minimizes the norm of this $\vec{\epsilon}$. To do this, we must use the SVD, which we will cover in week 3. For this project, all you need to know is that you need to run the command:

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[U, S, V] = svd(A);
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The vector h will then be the last column of V, and you can then construct the 3x3 homography matrix by reshaping the 9x1 h vector.